

Calculus Practice — Partial Derivatives and the Gradient

8 questions. Try all of them before checking the answers below.

Practice Questions

Q1 Find the gradient of $f(x, y) = e^{x^2-y^2}$.

Q2 Let $f(x, y) = x^2y - 3xy^2 + 7$. Find $\nabla f(x, y)$ and evaluate it at the point $(2, -1)$.

Q3 Let $f(x, y) = x^2 - y^2$. Find the critical point of f , and explain why it is neither a local maximum nor a local minimum.

Q4 Three cabins need to be connected to a single power station. The cabins sit at $(1, 2)$, $(4, 6)$, and $(7, 2)$ on a map (in km). Find the position (x, y) of the station that minimizes the total squared distance to all three cabins:

$$D(x, y) = (x - 1)^2 + (y - 2)^2 + (x - 4)^2 + (y - 6)^2 + (x - 7)^2 + (y - 2)^2$$

Q5 Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for:

$$f(x, y) = 5x^3y^2 - 4xy + 2y^4$$

Q6 Find all critical points of $f(x, y) = x^3 - 3xy + y^3$.

Q7 The temperature (in °C) at a point (x, y) inside a sauna is modeled by:

$$T(x, y) = 60 + 3(x - 2)^2 + (y - 4)^2$$

Find $\nabla T(x, y)$, and use it to find the coordinates of the coldest spot in the room and the temperature there.

Q8 Let $f(x, y) = x^2 + y^2 - 4x + 6y + 5$. Find the critical point of f , and the minimum value of f .

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Answers & Solutions

A1. Using the chain rule on each partial derivative:

$$\frac{\partial f}{\partial x} = 2x e^{x^2-y^2}, \quad \frac{\partial f}{\partial y} = -2y e^{x^2-y^2}$$

$$\nabla f(x, y) = (2x e^{x^2-y^2}, -2y e^{x^2-y^2})$$

A2. $\frac{\partial f}{\partial x} = 2xy - 3y^2$ and $\frac{\partial f}{\partial y} = x^2 - 6xy$, so:

$$\nabla f(x, y) = (2xy - 3y^2, x^2 - 6xy)$$

At $(2, -1)$: $\frac{\partial f}{\partial x} = 2(2)(-1) - 3(1) = -7$, and $\frac{\partial f}{\partial y} = 4 - 6(2)(-1) = 16$.

$$\nabla f(2, -1) = (-7, 16)$$

A3. $\nabla f = (2x, -2y) = (0, 0) \Rightarrow$ critical point at $(0, 0)$. Along the x -axis ($y = 0$), $f(x, 0) = x^2 \geq 0$, increasing as you move away from the origin — looks like a minimum. Along the y -axis ($x = 0$), $f(0, y) = -y^2 \leq 0$, decreasing as you move away — looks like a maximum. Since the behavior is a max in one direction and a min in another, $(0, 0)$ is a **saddle point**.

A4. $\frac{\partial D}{\partial x} = 2(x-1) + 2(x-4) + 2(x-7) = 6x - 24$, and $\frac{\partial D}{\partial y} = 2(y-2) + 2(y-6) + 2(y-2) = 6y - 20$. Setting both to zero:

$$x = 4, \quad y = \frac{20}{6} = \frac{10}{3}$$

So the optimal station location is $\left(4, \frac{10}{3}\right)$ — notice this is just the average of the cabins' x -coordinates and the average of their y -coordinates.

A5.

$$\frac{\partial f}{\partial x} = 15x^2y^2 - 4y, \quad \frac{\partial f}{\partial y} = 10x^3y - 4x + 8y^3$$

A6. $\frac{\partial f}{\partial x} = 3x^2 - 3y = 0 \Rightarrow y = x^2$, and $\frac{\partial f}{\partial y} = -3x + 3y^2 = 0 \Rightarrow x = y^2$. Substituting $y = x^2$ into $x = y^2$: $x = x^4 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0$ or $x = 1$.

- When $x = 0$: $y = 0$, giving critical point $(0, 0)$
- When $x = 1$: $y = 1$, giving critical point $(1, 1)$

A7. $\nabla T(x, y) = (6(x-2), 2(y-4))$. Setting both components to zero:

$$x = 2, \quad y = 4$$

The coldest spot is at $(2, 4)$, and since both squared terms vanish there, $T(2, 4) = 60^\circ\text{C}$.

A8. $\nabla f = (2x - 4, 2y + 6) = (0, 0) \Rightarrow$ critical point at $(2, -3)$. Completing the square confirms this is a minimum:

$$f(x, y) = (x - 2)^2 + (y + 3)^2 - 8$$

Since both squared terms are ≥ 0 , the minimum value is -8 , achieved at $(2, -3)$.